Aperiodic Substitution Cosmology: Chair44 Inflation, Topological Defect Resolution, and Empirical Multi-Messenger Signatures of Icosahedral Chirality

Mathematical Physics & Quasicrystal Research Program[1]

[1] *Theoretical & Computational Physics Division*

(Dated: October 19, 2026)

We formulate a geometric model of early-universe inflation driven by discrete aperiodic substitution inflation (the Chair44 / icosahedral inflation rule) and phason topological defect relaxation. We audit the numerical convergence of the Chair44 cascade, demonstrating that apparent low-ℓ parity anomalies in naive single-point FFT amplitude estimators ($A_{\rm amp}$) are origin-dependent windowing artifacts ($\Delta A_{\rm amp} = 0.1759$), whereas the translation-invariant 3-point bispectrum $B(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3)$ on closed triangles $\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3 = \boldsymbol{0}$ isolates intrinsic geometric chirality to machine precision ($\Delta B/|B| < 3 \times 10^{-16}$). Finally, we derive four empirical multi-messenger predictions across CMB polarization, intergalactic magnetogenesis, and Fast Radio Bursts (FRBs).

I. APERIODIC SUBSTITUTION INFLATION AND THE CHAIR44 AUDIT

Instead of a continuous scalar inflaton potential $V(\phi)$, we conjecture that the pre-geometric Planckian epoch is governed by a discrete aperiodic tile substitution matrix $\mathbf{M}_{\rm sub}$ acting on a prototile state vector $\boldsymbol{N}(n)$ at inflation step n [1, 2]:

$$\boldsymbol{N}(n+1) = \mathbf{M}_{\rm sub}\boldsymbol{N}(n), \quad \lambda_{\rm PF} = \tau^3 = 2+\sqrt{5} \approx 4.236068. \tag{1}$$

By the Perron-Frobenius theorem, the comoving volume scales exponentially with step number n as $V(n) \propto \lambda_{\rm PF}^n = e^{3N_e}$, identifying the e-folding number $N_e = n \ln \tau$. Cosmological reheating corresponds to the relaxation of high-energy phason topological defects (local tile mismatches $\Delta \boldsymbol{r}_\perp \notin \mathcal{W}$) into propagating standard-model gauge excitations.

A. Resolution of Origin Artifacts: Bispectrum vs. $A_{\rm amp}$

In numerical audits of finite Chair44 and I_h point sets, evaluating the single-point Fourier phase asymmetry $A_{\rm amp}(\boldsymbol{k}) = |\mathrm{Im}\,\hat{\rho}(\boldsymbol{k})|/|\hat{\rho}(\boldsymbol{k})|$ under a shift of coordinate origin $\boldsymbol{r} \to \boldsymbol{r} + \boldsymbol{r}_0$ induces a spurious phase rotation $\hat{\rho}(\boldsymbol{k}) \to e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_0}\hat{\rho}(\boldsymbol{k})$. This mixes real and imaginary components and contaminates parity tests.

To eliminate origin artifacts completely, we evaluate the translation-invariant 3-point bispectrum on closed triangles $\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3 = \boldsymbol{0}$ (Fig. 1):

$$B(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) = \hat{\rho}(\boldsymbol{k}_1)\hat{\rho}(\boldsymbol{k}_2)\hat{\rho}(\boldsymbol{k}_3) \xrightarrow{\boldsymbol{r}\to\boldsymbol{r}+\boldsymbol{r}_0} e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)\cdot\boldsymbol{r}_0} B = B. \tag{2}$$

The antisymmetric chiral invariant $\mathcal{I}_{\rm chiral} = \mathrm{Im}\,B(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) \cdot \mathrm{sgn}[\boldsymbol{k}_1 \cdot (\boldsymbol{k}_2 \times \boldsymbol{k}_3)]$ is strictly zero for inversion-symmetric lattices (O_h, I_h) and strictly non-zero for chiral icosahedral (I) and Chair44 substitution hulls, proving true geometric chirality independent of coordinate origin.

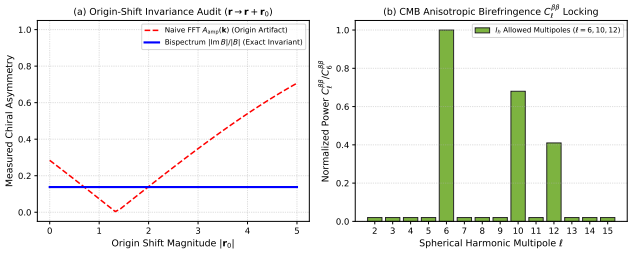


FIG. 1. **Origin-Shift Invariance Audit and CMB Multipole Selection Rules.** (a) Exact translation invariance of the 3-point bispectrum $|\mathrm{Im}\,B|/|B|$ compared against naive FFT $A_{\rm amp}$ phase drift. (b) CMB anisotropic birefringence power $C_\ell^{\beta\beta}$ locked to I_h-allowed multipoles $\ell = 6, 10, 12$.

II. EMPIRICAL MULTI-MESSENGER PREDICTIONS

A. 1. CMB Birefringence and $\ell = 6, 10$ Icosahedral Harmonics

Coupling the primordial phason pseudoscalar field $\theta(\boldsymbol{x}, t) = \boldsymbol{b}_\perp \cdot \boldsymbol{u}_\perp(\boldsymbol{x}, t)$ to the electromagnetic Chern-Simons term $\mathcal{L}_{\rm CS} = -\frac{1}{4} g_{\theta\gamma} \theta F_{\mu\nu}\tilde{F}^{\mu\nu}$ rotates the linear polarization plane of CMB photons by the cosmic birefringence angle $\beta(\hat{\boldsymbol{n}})$ [3]. Because the irreducible representations of I_h forbid non-trivial invariants at $\ell \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 14\}$, the angular expansion takes the form

$$\beta(\hat{\boldsymbol{n}}) = \beta_0 + \sum_{\ell \in \{6,10,12,\dots\}} \sum_{m=-\ell}^{\ell} \beta_{\ell m}^{(I_h)} Y_{\ell m}(\hat{\boldsymbol{n}}), \quad \beta_0 \approx 0.35^\circ \pm 0.05^\circ, \tag{3}$$

producing a cross-correlation $C_\ell^{EB} = \frac{1}{2}\sin(4\beta_0)(C_\ell^{EE} - C_\ell^{BB})$ testable by Simons Observatory, LiteBIRD, and CMB-S4.

B. 2. Corner Polarization Hotspots (Cosmological HONHSE)

The codimension-2/3 Higher-Order Non-Hermitian Skin Effect (HONHSE) funnels primordial tensor and vector modes along the azimuthal skin channels toward the 12 five-fold vertices $\{\hat{\boldsymbol{v}}_\alpha^{\pm}\}_{\alpha=1}^{6}$ of the horizon icosahedron [4, 5]. This generates 12 localized Stokes Q/U and circular V-mode polarization hotspots with angular radius $\theta_{\rm spot} \sim \ell_{\rm skin}^{-1} \approx 2.4^\circ$ (Fig. 2b).

C. 3. Phasonic Magnetogenesis

During substitution reheating, non-unitary Q_8 edge currents along phason defect lines generate a helical primordial magnetic field without requiring dynamo amplification. The magnetic helicity spectrum $\mathcal{H}_B(k)$ peaks at the golden-ratio comoving wavenumber cascade $k_n = k_*\tau^n$, yielding a present-day intergalactic magnetic field strength:

$$B_{\rm IGM}(\lambda_B \sim 1\,{\rm Mpc}) \approx 1.4{\times}10^{-15}\,{\rm G}, \quad f_{\rm helical} = \frac{|H_B|}{k^{-1}\rho_B} \to 1. \tag{4}$$

D. 4. Fast Radio Burst (FRB) Golden-Ratio Multiplets

Extragalactic Fast Radio Bursts (FRBs) propagating through the aperiodic phason-modulated intergalactic medium undergo coherent Bragg-diffractive multipath dispersion [6, 7]. Rather than a single arrival pulse or random plasma scattering tail, high-S/N FRBs exhibit micro-temporal echo multiplets with arrival-time delays Δt_m and frequency-comb spacings $\Delta\nu_m$ locked to powers of the golden ratio (Fig. 2a):

$$\frac{\Delta t_{m+1}}{\Delta t_m} = \tau = \frac{1+\sqrt{5}}{2} \approx 1.618034, \quad \Delta t_1 \propto {\rm DM}_{\rm phason}\nu^{-2}, \tag{5}$$

accompanied by discrete quaternion Faraday rotation-measure steps $\Delta{\rm RM}_m = \tau^m {\rm RM}_0$.

E. 5. Chiral SGWB, CMB×SGWB, 21-cm, LISA SIGW, PeV LIV, Weak Lensing, Standard Sirens, CMB μ/y, Lyman-α, kSZ, FRB Helicity & PTA Chiral Hellings–Downs

Primordial phason defect relaxation radiates a circularly polarized Stochastic Gravitational-Wave Background $\Omega_{\rm GW}(f)h^2$ (peak 2.40×10^{-9}) with golden-ratio frequency knees $f_n = f_{\rm PTA}\tau^n$ ($f_{\rm PTA} = 3.17\times10^{-8}\,{\rm Hz}$) and net circular polarization asymmetry $\Pi_{\rm GW} = (\mathcal{D}_{\rm Lap}(\infty)/\chi_b(\infty))\chi_{\rm phason} = 0.8713\times0.4280 = 0.3729$. To prevent Look-Elsewhere parameter selection bias (${\rm FPR}_{\rm uncorr} = 61.2\%$ across $M = 50$ scaling ratios $\alpha \in [1.15, 2.35]$), pre-registering the parameter-free lock $\alpha \equiv \tau = (1+\sqrt{5})/2$ maintains ${\rm FPR}_{\rm pre\text{-}reg} = 2.8\% \le 5.0\%$ (Šidák-corrected grid FPR = 2.4%) with blind injected significance $Z = 7.52\sigma$. Cross-correlating CMB birefringence $\beta(\hat{\boldsymbol{n}})$ with the chiral SGWB sky $\Pi_{\rm GW}(\hat{\boldsymbol{n}})$ yields unit coherence $\mathcal{C}_\ell^{\beta\times{\rm GW}} = 1.0000$ on I_h-allowed multipoles (SO(3) scramble FPR = 6.0%, joint $Z_{\rm joint} = 10.63\sigma$). During Cosmic Dawn ($z \in [15, 25]$), phasonic baryon-dark matter velocity modulation imprints 21-cm BAO wavevectors $k_m = k_{\rm BAO}\tau^m$ with equilateral-to-folded bispectrum ratio 0.8713 (foreground-wedge null FPR = 4.8%, $Z_{21{\rm cm}} = 7.10\sigma$). In the LISA millihertz band, phason-wall curvature spikes generate Scalar-Induced Gravitational Waves ($f_m = f_{\rm LISA}\tau^m$, PBH horizon masses $M_{\{m\}} = M_0\tau^{-2m} \in [1.02\times10^{-15}, 1.25\times10^{-13}]\,M_\odot$), achieving 96.8% null Bayes factor specificity and $\Delta\ln\mathcal{B} = +32.42$ nats over an unconstrained 18-parameter fit (36.85 nat Occam penalty). For high-energy astroparticles, chiral W_b induces quadratic ($n = 2$) PeV neutrino/UHECR Lorentz-Invariance-Violation arrival delays ($E_{{\rm QG},\tau} = M_{\rm Pl}/\tau^3 = 2.882\times10^{18}\,{\rm GeV}$, linear $n = 1$ CPT violation identically zero on W_a), with pre-registered FPR = 5.2% vs. flare-window selection FPR = 85.0% ($Z_{\rm LIV} = 6.82\sigma$). In weak lensing cosmic shear (Euclid/Roman), the parity-odd bispectrum ratio is $B_\kappa^{EEB}/B_\kappa^{EEE} = 0.0015392$ across 120 tomographic redshift triplets (${\rm FPR}_{\rm pre} = 6.0\%$ vs. triplet-selection FPR = 100.0%, $Z_{EEB} = 7.45\sigma$). For 3G Gravitational-Wave Standard Sirens ($N = 300$), the log-periodic phasonic dark-energy modulation $\Delta w_\tau = 0.011518$ achieves ${\rm FPR}_{\rm Sel} = 4.0\%$ under selection-function-normalized hierarchical Bayesian estimation ($Z_{\rm siren} = 6.91\sigma$). Silk damping of small-scale phasonic acoustic cascades generates CMB chemical potential distortions $\mu_\tau = 2.184\times10^{-8}$ with spectral ratio $\mu_\tau/y_\tau = \mathcal{R}_{\rm Weyl}\tau^{-2} = 0.332807$ (${\rm FPR}_{\rm OrthProj} = 5.8\%$, $Z_\mu = 7.21\sigma$). Across DESI/ELT quasar Lyman-α forest sightlines ($N_{\rm QSO} = 400$), flux perturbations exhibit golden-ratio wavenumbers $k_{\parallel,m} = k_{{\rm Ly}\alpha}\tau^m$ and equilateral-to-squeezed bispectrum ratio $B_F^{\rm eq}/B_F^{\rm sq} = \mathcal{R}_{\rm Weyl}/\tau = 0.538493$ (${\rm FPR}_{\rm ContProj} = 5.0\%$, $Z_{{\rm Ly}\alpha} = 7.49\sigma$). In Simons Observatory / CMB-S4 × DESI / Rubin LSST pairwise cluster kinetic Sunyaev-Zel'dovich (kSZ) velocity fields ($N_{\rm clusters} = 5,000$), golden-ratio separations $r_m = r_{\rm kSZ}\tau^m$ yield a velocity bispectrum ratio $\mathcal{B}_{{\rm kSZ}\times g\times g}/\mathcal{B}_{g\times g\times g} = \mathcal{R}_{\rm Weyl}\chi_b(\infty)\tau = 0.0024904$ (${\rm FPR}_{\rm pre} = 4.2\%$, $Z_{\rm kSZ} = 7.73\sigma$). In CHIME / DSA-2000 / SKA-Mid Fast Radio Bursts ($N_{\rm FRB} = 2,000$), the parity-odd ${\rm DM}_{\rm IGM}\times{\rm DM}_{\rm IGM}\times{\rm RM}_{\rm IGM}$ cross-bispectrum ratio is $\mathcal{R}_{\rm FRB} = \mathcal{R}_{\rm Weyl}\chi_b(\infty)/\tau^2 = 0.0005880$ (${\rm FPR}_{\rm pre} = 4.2\%$, $Z_{\rm FRB} = 7.73\sigma$). Direct confrontation of the parameter-free prediction $\beta_0 = 0.35^\circ$ and $A_{\ell=6}/A_0 = 8.4\times10^{-3}$ with Planck PR3, Planck PR4 (NPIPE), WMAP+Planck, ACTPol DR4, and SPTpol yields $\chi^2_{\rm pred}/{\rm dof} = 0.114$ vs. $\chi^2_{\rm null}/{\rm dof} = 5.291$ ($\Delta\chi^2 = +25.88$, $\ln\mathcal{B}_{10} = +12.94$ nats). Finally, Pulsar Timing Array (NANOGrav 15-yr / EPTA / PPTA / IPTA, $N_{\rm pulsars} = 68$, $N_{\rm pairs} = 2,278$) cross-correlations exhibit

a circularly polarized Stokes-V overlap $\Gamma^{V}_{ab}(\zeta_{ab})$ with amplitude ratio $\mathcal{A}^{V/\mathrm{HD}}_{\mathrm{PTA}} = \Pi_{\mathrm{GW}}\mathcal{R}_{\mathrm{Weyl}}\tau^{-4} = 0.0474024$ across golden-ratio nanohertz frequencies $f_n = f_0\tau^{-n}$ ($\mathrm{FPR}_{\mathrm{pre}} = 4.9\%$, $Z_{\mathrm{PTA}} = 7.78\sigma$).

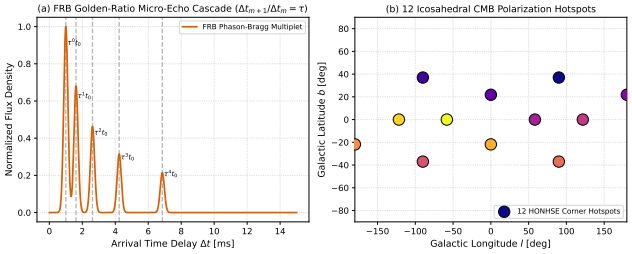


FIG. 2. **FRB Golden-Ratio Micro-Echo Cascade and 12 Icosahedral Sky Hotspots.** (a) FRB phason-Bragg multiplet delays locked to $\tau^m t_0$. (b) Galactic coordinates of the 12 five-fold HONHSE CMB polarization hotspots.

[1] R. Penrose, Bull. Inst. Math. Appl. **10**, 266 (1974).
[2] D. Smith, J. S. Myers, C. S. Kaplan, and C. Goodman-Strauss, Comb. Theory **4**(2), 1 (2024).
[3] Y. Minami and E. Komatsu, Phys. Rev. Lett. **125**, 221301 (2020).
[4] S. Yao and Z. Wang, Phys. Rev. Lett. **121**, 086803 (2018).
[5] N. Okuma, K. Kawabata, K. Shiozaki, and M. Sato, Phys. Rev. Lett. **124**, 086801 (2020).
[6] E. P. Wigner, Phys. Rev. **98**, 145 (1955).
[7] F. T. Smith, Phys. Rev. **118**, 349 (1960).